From (1), $RT - \frac{4}{5}RT = \frac{4}{10}T$. Hence $\frac{1}{5}RT = \frac{2}{5}T \Longrightarrow RT = 2T$.

From Equation (2), $RT = (2 - \frac{1}{2})(\frac{RT}{2} - \frac{5}{2}) = \frac{3}{2}(\frac{RT+5}{2}) = \frac{3}{4}(RT + 5)$.
Thus $RT = 15$. The distance travelled is 15km.

Qn14. The locker labelling requires $\frac{1390600}{200} = 6953$. Lockers 1 through 9 require 9 digits. Lockers 10 through 99 require $2 \times 90 = 180$ digits. Lockers 100 through 999 require $3 \times 900 = 2700$ digits. Hence the remaining lockers require $6953 - 2700 - 180 - 9 = 4064$ digits, and so there must be $\frac{4064}{4} = 1016$ more lockers each using 4 digits. In all, there are $1016 + 999 = 2015$ student lockers.

Qn15. Let N be the centre of the circle and then draw the two radii NP and NQ so that $AN = NB = NP = NQ = 6cm$. $\triangle ABC$ is equilateral so $< PAN = 60^\circ$ and hence $< APN = 60^\circ$ since $AN = NP$. It follows that $< ANP = 60^\circ$ so, $\triangle ANP$ is equilateral.

Similarly, $< QNB = 60^\circ$ and therefore $< PNQ = 60^\circ$ as well. Let $\bigotimes$ be the area of the figure bounded by the points. Then

$$\begin{aligned} \otimes ABC &= \otimes ANP + \otimes BNQ + \otimes NPQ + \otimes PQC \\ \otimes PQC &= \otimes ABC - (\otimes ANP + \otimes BNQ + \otimes NPQ) \end{aligned}$$

$$\text{But} \otimes ANP = \frac{1}{2} \times a \times b \times \sin\theta = \frac{1}{2} \times 6 \times 6 \times \sin 60^\circ = 9\sqrt{3}.$$

$$\text{Similary} \otimes BNQ = \frac{1}{2} \times a \times b \times \sin\theta = \frac{1}{2} \times 6 \times 6 \times \sin 60^\circ = 9\sqrt{3}.$$

$$\text{Also} \otimes NPQ = \frac{\theta}{360} \times \pi r^2 = \frac{60}{360} \times \pi \times 6^2 = 6\pi.$$

$$\otimes ABC = \frac{1}{2} \times 12 \times 12 \times \sin 60^\circ = \frac{72\sqrt{3}}{2} = 36\sqrt{3}.$$

$$\text{Hence} \otimes PQC = 36\sqrt{3} - 9\sqrt{3} - 9\sqrt{3} - 6\pi = 18\sqrt{3} - 6\pi = 6(3\sqrt{3} - \pi)cm^2.$$

NMC 2015 O-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. For what value of k is the displayed factoring valid?

$$2015^3 + 2015^4 = k(1 + 2015).$$

Qn2. When the decimal point of a certain positive decimal number is moved four places to the right, the new number is nine times the reciprocal of the original number. What is the original number?

Qn3. A line with slope 2 intersects a line with slope 5 at the point $(10, 20)$. What is the distance between the x-intercepts of these two lines?

Qn4. For what value(s) of x is $8xy - 12y + 2x - 3 = 0$ true for all values of y?

Qn5. An employee received a salary raise of 2% last year and 3% this year. After these raises, her current salary is UGX 651, 372. What was her salary two years ago (before the first of the two salary raises).

Qn6. Alex, Barbra and Charity share $Shs1,000,000$ among themselves not necessarily getting equal amounts, to be invested in different ways. At the end of one year they have a total of $Shs1,500,000$. Barbra and Charity have both doubled their money, where as Alex has lost $Shs100,000$. What was Alex's original portion?

Qn7. Let g be the function such that $g\left(\frac{x}{3}\right) = x^2 + x + 1$. Find all values of z for which $g(3z) = 7$.

Qn8. Two numbers are such that their difference, their sum and their product are in the ratio $1 : 7 : 24$ respectively. Find the product of these two numbers.

Qn9. Suppose A, B and C are three numbers for which $1001C - 2002A = 4004$ and $1001B + 3003A = 5005$. What is the average of the three numbers A, B and C?

Qn10. A company sells milk in cylindrical jars. Marketing research suggests that using wider jars will increase sales. Suppose that the diameter of the jars is increased by 25% without altering the volume. By what percentage must the height be decreased?

SECTION B

Qn11. For all real x, $f(x + 1) + f(x) = 1$. If $f(9) = 18$. What is the value of $f(1) + 2$?

Qn12. (a) Let f be the function defined by $f(x) = ax^2 - \sqrt{2}$ for some positive a. If $f(f(\sqrt{2})) = -\sqrt{2}$, what is the value of a?

(b) When students sat in a dinning hall for lunch each on a separate table, there was one person left. As a result they decided to sit two per table and one table was left free. How many tables were in the dinning hall?

Qn13. (a) Country A has c% of the world's population and owns d% of the world's wealth. Country B has e% of the world's population and f% of its wealth. Assume that the citizens of A share the wealth of A equally, and assume that those of B share the wealth of B equally. Find the ratio of the wealth of a citizen of A to the wealth of a citizen of B.

(b) Some boys and girls are having a car wash to raise money for a class trip to China. Initially 40% of the group are girls. Shortly thereafter two girls leave and two boys arrive, and then 30% of the group are girls. How many girls were initially in the group?

Qn14. (a) One of the solutions of the equations $(a - w)(a - x)(a - y)(a - z) = 9$ is $a = 2$. If w, x, y, z are different integers, find the value of $w + x + y + z$.

(b) Two non-zero real numbers x and y satisfy the relation $xy = x - y$. Find the value of $\frac{x}{y} + \frac{y}{x} - xy$.

Qn15. Point P is inside an equilateral triangle $\triangle ABC$. Points Q, R and S are the feet of the perpendicular from P to $\overline{AB}, \overline{BC}$ and $\overline{CA}$ respectively. Given that $\overline{PQ} = 1, \overline{PR} = 2$ and $\overline{PS} = 3$, what is $\overline{AB}$?

NMC 2015 O-LEVEL PAPER 2, SOLUTIONS

SECTION A

Qn1. $2015^3 + 2015^4 = 1015^3(1 + 2015)$. So, by comparing with $k(1 + 2015)$ we have $k = 2015^3$.

Qn2. Let the number be y. By moving the decimal point four places to the right, it is the same as multiplying the by 10000. Hence $10000y = 9(\frac{1}{y}) \implies 10000y^2 = 9$.

Therefore $y = \sqrt{\frac{9}{10000}} = \frac{3}{100} = 0.03$.

Qn3. The equation of a line is given by $y = mx + c$. For the line of slope 2 that passes through the point $(10, 20)$, its equation is given by $\frac{y-20}{x-10} = 2 \implies y = 2x$. Similarly, the equation of

the line of slope 5 that passes through the point $(10, 20)$, is $\frac{y-20}{x-10} = 5 \implies y = 5x - 30$. The $x-$intercepts are obtained by setting $y = 0$ in the respective equations, hence $x = 0$ and $x = 6$. Therefore the distance between the points is $6 - 0 = 6$.

Qn4.

$$\begin{aligned} 8xy - 12y + 2x - 3 &= 0 \\ 4y(2x - 3) + 2x - 3 &= 0 \\ (2x - 3)(4y + 1) &= 0 \\ x = 1.5 \text{ or } y &= -0.25. \end{aligned}$$

Qn5. Let the original salary be x. Then $(1.02x) \times (1.03) = 651372 \implies 1.0586x = 651372$ or $x = \frac{651372}{1.0586} = 620,000$.

Qn6. Denote the original portions for Alex, Betty, and Charity as a, b, and c respectively. Then $a+b+c = 1000000$ and $a-100000+2(b+c) = 1500000$. Substituting $b+c = 1000000-a$ in the second equation, we have $a-100000+2(1000000-a) = 1500000$. This yields $a = Shs.\ 400,000$ which is Alex's original portion.

Qn7. Let $x = 9z$, then $g\left(\frac{9z}{3}\right) = 81z^2 + 9z + 1$. This implies that $81z^2 + 9z + 1 = 7$. Solving this equation by factorization method, we have $3(3z + 1)(9z + 2) = 0$. Thus $z = -\frac{1}{3}$ or $z = \frac{2}{9}$.

Qn8. Let the numbers be represented by x and y. Then $x + y = 7(x - y) \implies 8y = 6x$.

Also, $xy = 24(x - y) \implies xy = 24x - 24y = 24x - 18x = 6x \implies x(y - 6) = 0$.

Since $x \neq 0$, it implies $y = 6$. Also since $6x = 8y \implies x = 8$. Thus $xy = 48$.

Qn9. Adding the two equations gives $1001A + 1001B + 1001C = 9009$. Hence

$$\begin{aligned} 1001(A+B+C) &= 9009 \\ A+B+C &= 9 \\ \\ \frac{A+B+C}{3} &= 3. \end{aligned}$$

The average of the three numbers A, B and C is 3.

Qn10. Let r, h and v respectively be the radius, height and volume of the jar that is currently being used. Then the new jar will have the radius of $1.25r$ and the volume is maintained as v. If H is the height of the new jar, then $v = \pi r^2 h = \pi(1.25r)^2 H \Longrightarrow \frac{H}{h} = \frac{1}{(1.25)^2} = 0.64$. Therefore H is 64% of h, so the height of a old jar must be reduced by $(100 - 64)\% = 36\%$

SECTION B

Qn11.

$$\begin{aligned} f(9)+f(8) &= 1 \Longrightarrow f(8) = 1 - f(9). \\ f(8)+f(7) &= 1 \Longrightarrow f(7) = 1 - f(8) = 1 - 1 + f(9) = f(9). \\ f(7)+f(6) &= 1 \Longrightarrow f(6) = 1 - f(7) = 1 - f(9). \\ f(6)+f(5) &= 1 \Longrightarrow f(5) = 1 - f(6) = 1 - 1 + f(9) = f(9). \\ f(5)+f(4) &= 1 \Longrightarrow f(4) = 1 - f(5) = 1 - f(9). \\ f(4)+f(3) &= 1 \Longrightarrow f(3) = 1 - f(4) = 1 - f(9) = 1 - 1 + f(9) = f(9). \\ f(3)+f(2) &= 1 \Longrightarrow f(2) = 1 - f(3) = 1 - f(9). \\ f(2)+f(1) &= 1 \Longrightarrow f(1) = 1 - f(2) = 1 - 1 + f(9) = f(9). \end{aligned}$$

Therefore $f(1) + 2 = f(9) + 2 = 18 + 2 = 20$.

Qn12. (a) Since $a \neq 0$, the only x for which $f(x) = -\sqrt{2}$ is $x = 0$. Since $f(f(\sqrt{2})) = -\sqrt{2}$, then $f(\sqrt{2})$ must be zero. Finally $f(\sqrt{2}) = 0 \Longrightarrow a(\sqrt{2})^2 - \sqrt{2} = 0$. Hence $a = \frac{\sqrt{2}}{2}$.

ALTERNATIVELY

$f(f(\sqrt{2})) = f(2a - \sqrt{2}) = a(2a - \sqrt{2})^2 - \sqrt{2} = -\sqrt{2}$.

Hence $a(2a - \sqrt{2})^2 = 0 \Longrightarrow 2a - \sqrt{2} = 0$ so that $a = \frac{\sqrt{2}}{2}$.

(b) Let s be the number of students and t the number of tables. Then $s = t + 1$ and $s = 2(t - 1)$. Since the two equations are equal, we have $t + 1 = 2t - 2 \Longrightarrow t = 3$. Thus, there are 3 tables in the dining hall.

Qn13. (a) Country A has $c\%$ of the world's population and owns $d\%$ of the world's wealth. Let P be the world's population and W its wealth. Then $\frac{Pc}{100}$ citizens of A together own $\frac{Wd}{100}$ units of wealth. So, each citizen of A owns $\frac{Wd}{100} \div \frac{Pc}{100} = \frac{Wd}{100} \times \frac{100}{Pc} = \frac{Wd}{Pc}$ units of wealth.

Similarly, each citizen of B owns $\frac{Wf}{100} \div \frac{Pe}{100} = \frac{Wf}{100} \times \frac{100}{Pe} = \frac{Wf}{Pe}$ units of wealth.

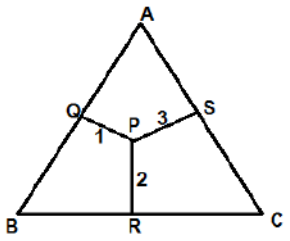
The required ratio is thus $\frac{Wd}{Pc} \div \frac{Wf}{Pe} = \frac{Wd}{Pc} \times \frac{Pe}{Wf} = \frac{de}{cf}$.

(b) Let g be the number of girls and b be the number of boys initially in the group. Then $g = 0.4(g + b)$.
After the two girls leave and two boys arrive, the size of the entire group is unchanged. So $g - 2 = 0.3(g + b)$.
Solving these two equations simultaneously, gives $g - 2 = 0.3 \times \frac{g}{0.4} = 0.75g$.
This implies $(1 - 0.75)g = 2 \Longrightarrow g = 8$.
Thus $b = \frac{0.6g}{0.4} = \frac{0.6\times 8}{0.4} = \frac{4.8}{0.4} = 12$. Therefore, initially we had 8 girls in the group.

Qn14. (a) Since $a = 2$ is a soluion to the equation, then the product $(2-w)(2-x)(2-y)(2-z) = 9$. Since $w \neq x \neq y \neq z$, all the four factors on the left side must have different values. i.e $(2 - w)(2 - x)(2 - y)(2 - z) = 3 \times -3 \times 1 \times -1$ not necessarily in the same order. Thus $2-w+2-x+2-y+2-z = 3-3+1-1 = 0 \Longrightarrow 8-(w+x+y+z) = 0 \Longrightarrow w+x+y+z = 8$.

(b) $\frac{x}{y} + \frac{y}{x} - xy = \frac{x^2+y^2-(xy)^2}{xy} = \frac{x^2+y^2-(x-y)^2}{xy} = \frac{x^2-y^2-x^2+2xy-y^2}{xy} = \frac{2xy}{xy} = 2$.

Qn15. Let the side length of $\triangle ABC$ be s. Then, the area of $\triangle APB = \frac{1}{2}bh = \frac{1}{2} \times s \times 1 = \frac{1}{2}s$.
The area of $\triangle BPC = \frac{1}{2}bh = \frac{1}{2} \times s \times 2 = s$ and the area of $\triangle CPA = \frac{1}{2}bh = \frac{1}{2} \times s \times 3 = \frac{3s}{2}$.



The area of $\triangle ABC$ is the sum of these three areas. Therefore area of $\triangle ABC = \frac{1}{2} \times s \times \frac{\sqrt{3}}{2} =$

$\frac{\sqrt{3}}{4}s^2 = s + \frac{1}{2}s + \frac{3s}{2} = 3s$. Hence $3s = \frac{\sqrt{3}}{4}s^2 \Longrightarrow (12 - \sqrt{3}s)s = 0$.

Since $s \neq 0$, then $12 - \sqrt{3}s = 0 \Longrightarrow s = \frac{12}{\sqrt{3}} = 4\sqrt{3}$.